

- 4 The Grand Canyon is located in Arizona, USA. It is estimated to have formed over five million years ago as the Colorado River began to create a deep channel (canyon) in the surrounding rocks.

Before the canyon formed, an ancestral species of antelope squirrel lived in the area. An antelope squirrel is a type of rodent and member of the squirrel family, Sciuridae.

It is estimated that around 3.6 million years ago, an ancestral species diverged into the two species that are present today.

- Harris's antelope squirrel, *Ammospermophilus harrisi*, has its habitat range extending from the south rim of the canyon.
- The white-tailed antelope squirrel, *Ammospermophilus leucurus*, has its habitat range extending from the north rim of the canyon.

Fig. 4.1 shows the location of the Colorado River in the Grand Canyon and the location of these species of antelope squirrel.

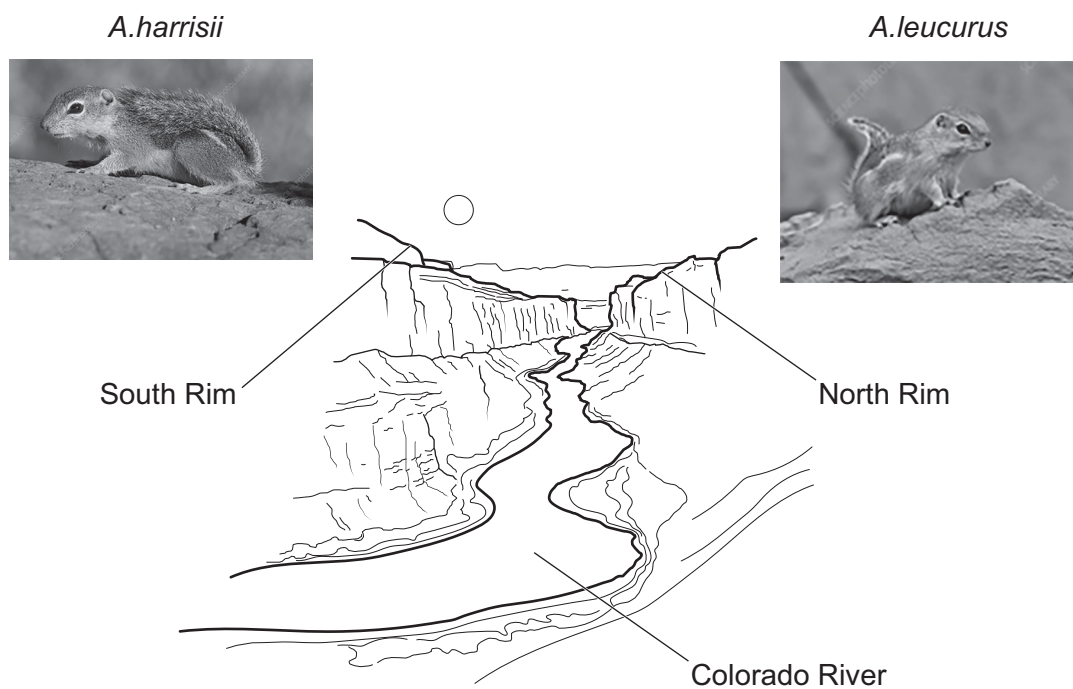


Fig. 4.1

- (a) Suggest **and** explain how *A. harrisii* and *A. leucurus* evolved from an ancestral species.

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- (b) Scientists investigated the evolutionary relationships of the squirrel family, Sciuridae. The scientists took samples from the current species in the family and carried out DNA sequencing and morphological analysis.

To compare current species with species from the past:

- specimens from museums were used to provide the tissue for DNA sequencing
- teeth and skulls from fossils were compared as part of the morphological analysis.

Describe the advantages of using DNA sequencing rather than morphological analysis to find out more about the evolutionary relationships of the squirrel family.

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- (c) The estimate for the date that an ancestral species diverged into *A. harrisii* and *A. leucurus* is 3.58 million years ago. This estimate has a large uncertainty.

DNA sequencing, including DNA from fossils, was used to estimate this date of divergence.

Suggest a reason why there is such a large uncertainty for this date estimate.

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[Total: 8]